Project Proposal

Numerical Simulation of Manufacturing Processes

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Triply Periodic Minimal Surface (TPMS)

TMPS are surfaces with the following mathematical properties [Wik25]:

- Nearly all studied TPMS are free of self-intersections (i.e. embedded in $\mathbb{R}^3$).
- All connected TPMS have genus ≥ 3, and in every lattice there exist orientable embedded TPMS of every genus ≥ 3.
- Embedded TPMS are orientable and divide space into two disjoint sub-volumes (labyrinths).

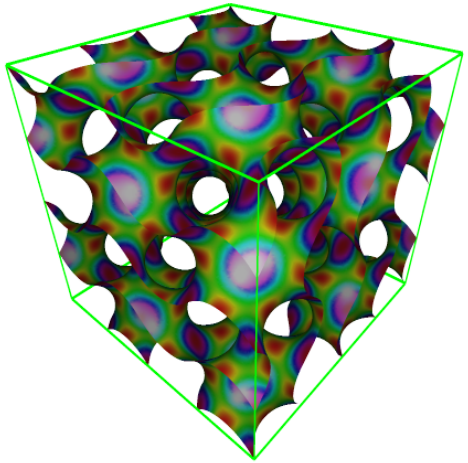
The mathematical background of TPMS is not particularly relevant for this project, but we will make use of the last property.

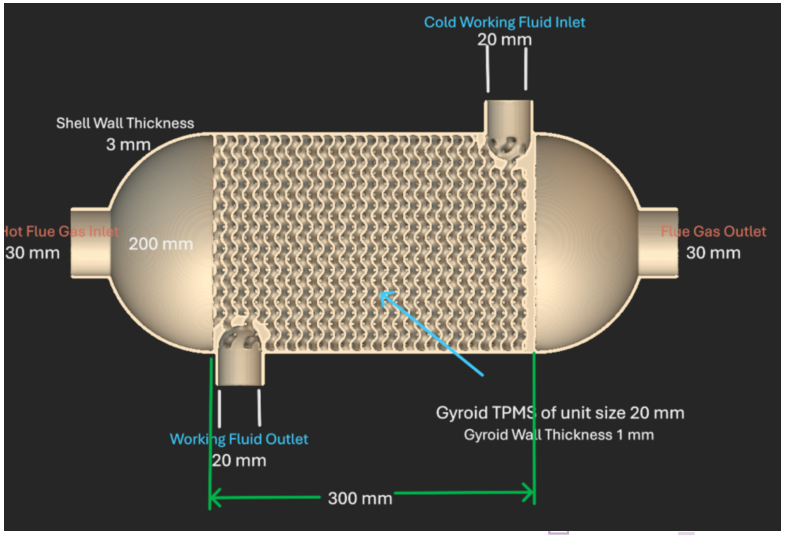
Gyroid-TPMS

Discovered by NASA scientist Alan Schoen in 1970, proved to be embedded by Grosse-Brauckmann and Wohlgemuth in 1996 [Wik26]. It is the structure that we will be treating in this project.

- Embedded; divides space into two *disjoint* sub-volumes.
- Can be approximated a the relatively simple equation

$$\sin x \cos y + \sin y \cos z + \sin z \cos x = 0$$





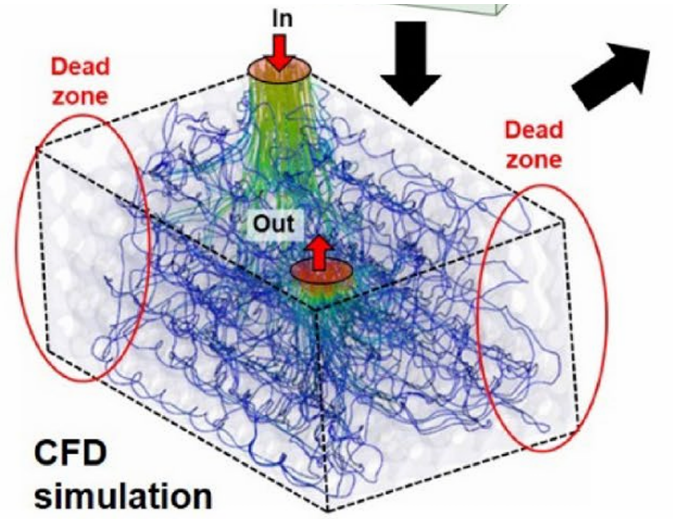
Advantages

TPMS provides a compelling alternative to traditional fin-based heat exchangers.

- High surface-to-volume ratio improves heat exchange.
- Continous TPMS walls generally provide good structural integrity (high stiffness).
- Geometry can be fine by function-based parameters.

The Problem

Dead zones appear in the structure. This causes poor flow utilization, which negatively affects the performance of the heat exchanger.



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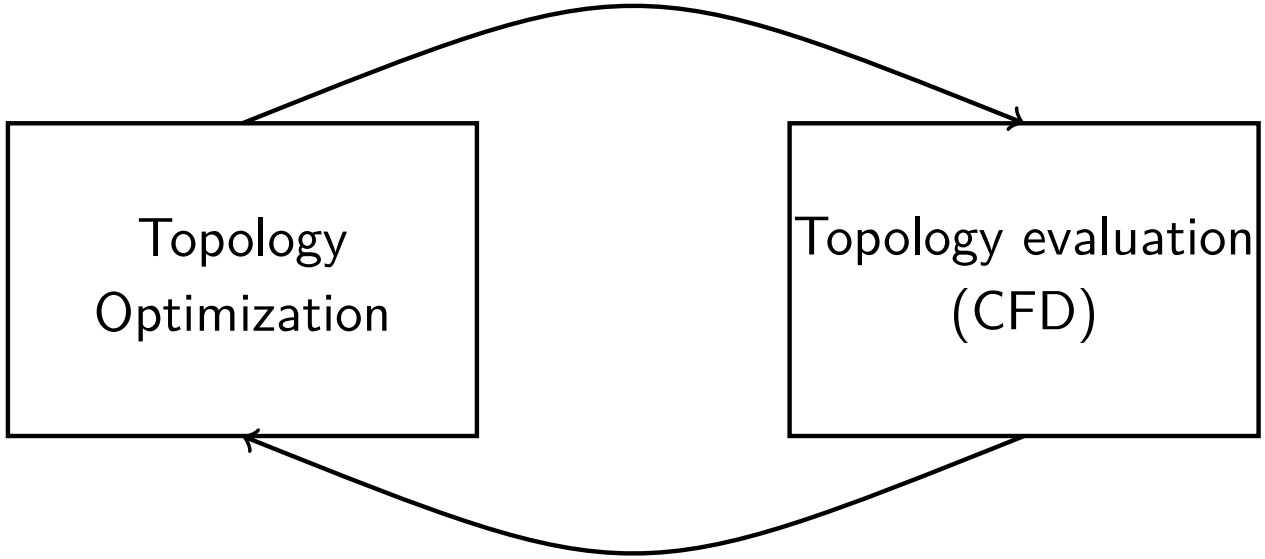
The Solution: Topology Optimization



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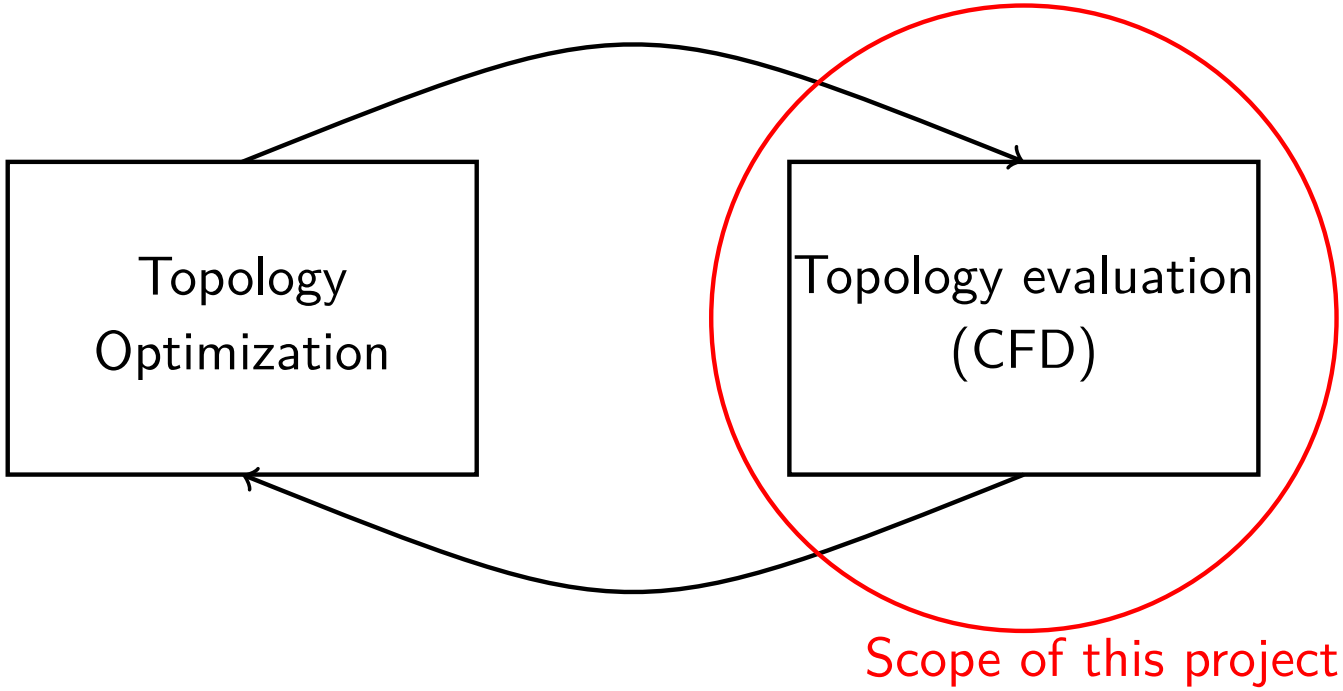

Current Methods

There are several ways to evaluate the performance of a given heat exchanger geometry.

- *Body-fitted Mesh* is the 'traditional' method which involves meshing of the structure, and evaluation in a FVM framework.
  - Well documented but computationally expensive for solving and re-meshing.
- *Equivalent porous medium* replaces the TMPS with averaged porous material.
  - Very efficient but relatively poor performance [PBE25]
- *Brinkman penalization* is the method of choice as it can provide sufficient precision in combination with relatively fast results.

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How it works

- The solid regions are treated as porous regions, with permeability $\eta \to 0$ (approaching solid).
- A penalty term proportional to the Brinkman force is then added to the Navier-Stokes equation.
- The penalty term is chosen in such a way that the momentum of the fluids approaches zero inside the solid region, making it effectively non-permeable.

Why it works

- 'Traditional' CFD methods require geometry-specific meshes to be generated. This process can be computationally expensive, especially for complex (read TPMS) geometries.
- During topology optimization, the need to re-mesh the geometry at every iteration renders 'traditional' methods practically unusable for TPMS geometries.
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[PBE25] A.I. Popov, D.M. Bragin, and A.V. Eremin.
Heat and mass transfer in tpms-based porous media: Two-medium approach.
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